Qassim University · College of Computer · Computer Science Department

Artificial Intelligence Attendance System

[AIAS]

Qassim University | College of Computer | جامعة القصيم كلية الحاسب

CS 499 – Graduation Project | 2025/2026 Supervisor: **Dr. Boulbaba Benammar** Department of Computer Science

📌Introduction

Attendance monitoring is a fundamental activity in academic institutions and plays a crucial role in evaluating student participation and performance. Traditional methods — such as calling names or passing attendance sheets — are time-consuming, prone to human error, and vulnerable to proxy attendance. With the advancement of Artificial Intelligence and computer vision, face-based attendance systems have become a practical solution. AIAS is an AI-powered Windows desktop application connected to a ceiling-mounted camera that captures periodic images of students during the lecture. Using YOLO for face detection and ArcFace/FaceNet for recognition, the system identifies students and updates attendance automatically throughout the session.

⚠️ Problem Statement

Manual attendance in university classrooms introduces several critical challenges:

- Time-consuming process that interrupts the lecture flow and reduces effective teaching time
- Proxy attendance fraud students sign for absent peers, compromising academic integrity
- Human errors and inconsistencies in large classrooms with many students
- Instructors must manually organize, digitize, and store attendance records
- Existing automated systems require close-range interaction or manual confirmation

🎯Aim & Objectives

Aim: To develop an intelligent AI-based system that accurately detects student presence and automates the entire attendance-tracking process.

- Design a Windows desktop application for automated multi-face detection and recognition
- Update attendance every 5 minutes to capture late arrivals and early departures
- Generate and email attendance reports automatically after each lecture session
- Ensure secure storage of biometric and attendance data via MongoDB
- Eliminate proxy attendance and reduce instructor workload entirely

💡Proposed Solution

AIAS operates as a fully automated, non-interactive pipeline. The instructor logs in, selects the course, and the system handles everything autonomously without any manual input during the lecture.

📷 **Capture** Camera every 5 min → 🔍 **Detect** YOLO multi-face → 🧠 **Recognize** ArcFace embeddings → 🗄 **Match** MongoDB database → 📊 **Classify** Update status → ✉️ **Report** Email instructor

🛠️Development Tools

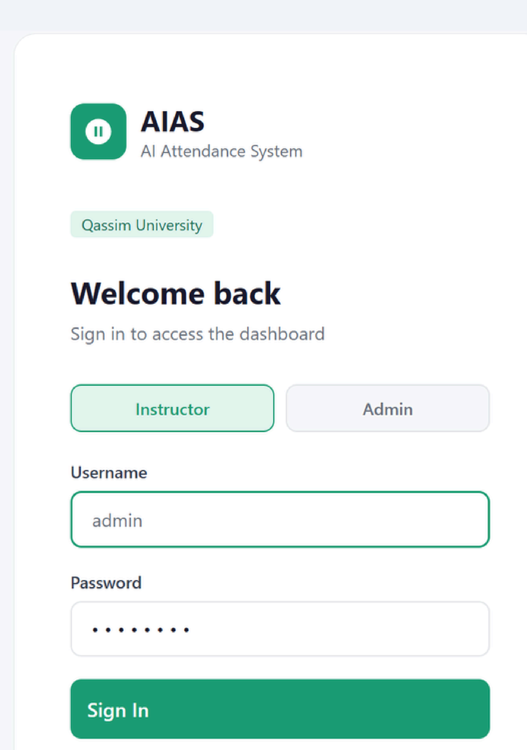
- **Python** – Main programming language
- **YOLO** – Multi-face detection
- **OpenCV** – Image processing & camera
- **MongoDB** – NoSQL database
- **PyQt5** – Desktop GUI (Windows)

📈 Results

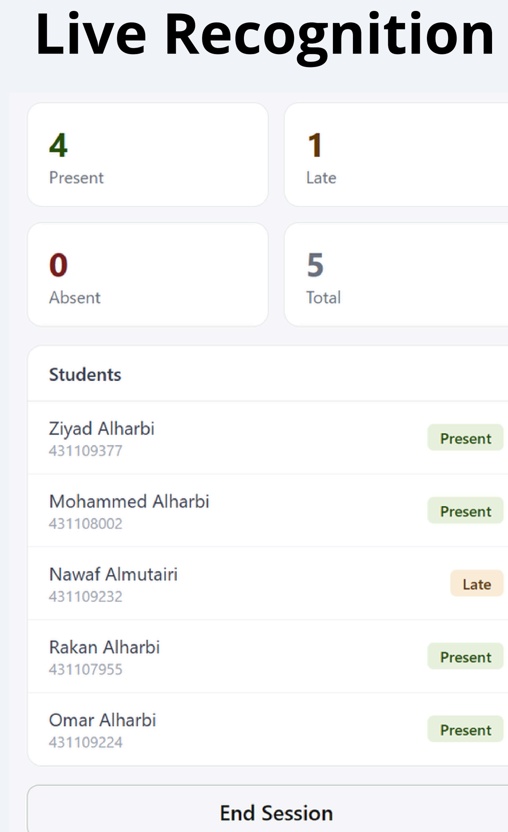
The system classifies each student's attendance status automatically based on predefined time rules:

Status	Rule	Description
✓ Present	Seen within first 10 min	Student arrived on time
⏱ Late	First seen after 10 min	Student arrived after grace period
✗ Absent	Never recognized	Not detected during lecture
⇠ Early Leave	Not seen 15+ min before end	Left before lecture ended

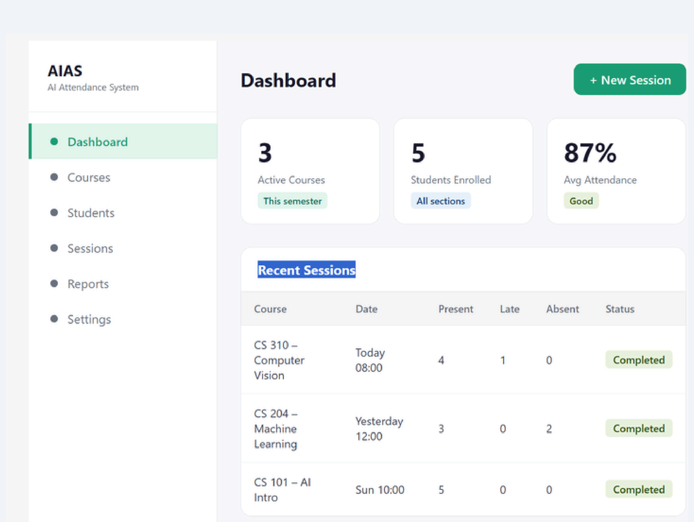
Admin Panel



Live Recognition



Instructor Dashboard



STUDENTS

- Mohammed Alharbi 431108002
- Ziyad Alharbi 431109377
- Nawaf Almutairi 431109232
- Rakan Alharbi 431107955
- Omar Alharbi 431109224

SUPERVISED BY
Dr. Boulbaba Benammar
Computer Science Department